

Chapter 2: Design Artifacts and Analysis

Farm Supply Inventory System

VI. Wireframes

The actual Farm Supply Inventory System uses a shared application shell with a left sidebar, a top header, and role-based views. After login, the first screen depends on the user role: **customer** users land on **Customer Home**, **staff** users land on **Staff Desk**, and **manager**, **admin**, and **super_admin** users land on **Dashboard**. Because the PDF requires a dashboard wireframe, the first diagram below follows the real **Dashboard** page used by manager, admin, and super admin users in the current system.

A. Dashboard Page Wireframe

Representative screen: dashboardView

```
+-----+
| SIDEBAR                                     | TOP HEADER
| Farm Supply Inventory                     | Dashboard
| Daily inventory workspace                 | See stock, sales,
| [Control Center pill]                   | User box | Refres
|
| Dashboard
| Staff Desk (role-based)
| Home / Shop / My Orders / Favorites / Profile (customer-only)
| POS / POS Counter (role-based)
| Operations
| Reports / Shift Reports (role-based)
| Users (admin+)
| Super Admin (super_admin only)
+-----+
| ADMIN OVERVIEW / WORKSPACE GUIDE
| Check today's priorities first.
| [Open Stock Tasks] [Open Reports]
| [1. Watch] [2. Update] [3. Review]
+-----+
| KPI CARDS
| Products | Active Products | Low Stock | Out Of Stock | Inventory Value | Today Sales | Mon
| Branches
+-----+
| Low-Stock Watch                         | Recent Activity
| Product | Stock | Reorder | Status      | transaction number | type | product |
+-----+
| Top Movers                             | Stock by Category      | Monthly
```

Product	Moved Qty	Category	Stock	Month
+-----				

- 1. The left sidebar is the shared navigation shell of the system, and some menu items only appear for specific roles.
- 2. The top header shows the active view title, subtitle, current user, and the Refresh and Logout controls.
- 3. The dashboard starts with the real Admin overview guide panel and its shortcut buttons.
- 4. The workflow strip matches the actual dashboard guidance in the system: Watch, Update, and Review.
- 5. The KPI area reflects the metrics loaded by the real dashboard API: products, active products, low stock, out of stock, inventory value, today sales, month sales, suppliers, and branches.
- 6. The Low-Stock Watch table lists products below reorder point.
- 7. The Recent Activity panel shows recent stock transactions.
- 8. The lower analytics area contains Top Movers, Stock by Category, and Monthly Sales.

B. Main Transaction Page Wireframe

Representative transaction: POS Counter / Point of Sale

The current system contains several transactions, but the clearest dedicated transaction screen in the actual UI is the POS page. It is directly connected to the Staff Desk through the Open POS action.

+-----	
TOP HEADER	
Point of Sale Process sales and review recent counter activity.	
User box Refresh Logout	
+-----	
POS HERO / STEP STRIP	
Fast counter checkout	
Warehouse label + live stock source	
[1 Select warehouse] [2 Add products] [3 Complete payment]	
Catalog items Available units Cart items Grand total	
+-----	
1. PICK PRODUCTS	2. REVIEW AND PAY
Warehouse select	Customer name (optional)
Search product / SKU / category	Payment method: Cash / GCash
Apply Filter / Clear Search	Discount type: None / Senior /
Live product cards with:	Discount amount
- product name	Senior/PWD ID number (condition
- SKU and unit	Senior/PWD proof upload (condit
- stock badge	Amount tendered (cash)
- price	Cash quick buttons: Exact / +10

```

| - available units | GCash reference number (conditi
| - reorder point | GCash receipt upload (condition
| - batch / expiry info | Notes
| - qty input + Add to Sale | Sale cart with editable line qu
| | Items | Subtotal | Discount | G
| | Amount tendered | Balance due |
| | Payment status | Complete Sale
+-----+
| 3. SALE RESULT / RECENT POS SALES
| Success summary card after checkout
| Print Last Receipt
| Recent POS Sales table: Sale # | Cashier | Items | Total | Payment | GCash Ref | Date
+-----+

```

1. The POS page begins with a dedicated POS Workspace hero and step strip, not with a plain form.
2. The warehouse selector is required because the catalog is warehouse-aware and loads live stock from the chosen source.
3. Product cards show quantity, stock, price, reorder point, and batch or expiry context before items are added to the sale.
4. The payment area changes dynamically based on the selected payment method and discount type.
5. Cash mode shows Amount Tendered and quick-cash buttons, while GCash mode shows reference and receipt upload fields.
6. Senior and PWD discounts reveal ID number and ID proof inputs because the real system supports verified discount handling.
7. The cart is editable directly from the checkout panel, and the total summary includes subtotal, discount, grand total, balance due, change, and payment status.
8. After a successful sale, the result appears in the Sale result card and can be printed through Print Last Receipt.

VII. Task Analysis

Main transaction analyzed: Staff POS checkout

Step	User Action	System Response
1	A staff user opens the system and logs in.	The system authenticates the account and opens Staff Desk as the default staff landing view.
2	The user clicks Open POS from Staff Desk , or opens POS Counter from the sidebar.	The system switches to the Point of Sale view.
3	The user selects a warehouse.	The system loads the live POS catalog, warehouse label, stock-source note, and POS header metrics.
4	The user searches the catalog and enters item quantities.	The system filters product cards and shows availability, reorder point, batch count, and expiry information.
5	The user clicks Add to Sale on one or more products.	The system adds products to the sale cart and updates items, subtotal, discount, grand total, and payment status.
6	The user enters optional customer name, selects payment method, and chooses a discount type.	The system reveals the correct conditional fields such as cash tendered, GCash reference and receipt upload, or senior/PWD verification inputs.
7	The user reviews the cart and adjusts quantities if needed.	The system recalculates subtotal, discount, amount tendered, balance due, and change in real time.
8	The user clicks Complete Sale .	Laravel validation checks request fields, required proof inputs, payment readiness, and stock availability in the selected warehouse.
9	If the transaction is valid, the system stores the sale.	The system creates sale records, sale items, related stock-out transactions, and deducts inventory from the correct warehouse batches.
10	The transaction finishes successfully.	The system shows the success summary in Sale result , adds the sale to Recent POS Sales , and enables Print Last Receipt .

VIII. Gantt Chart

The schedule below presents a practical development timeline covering planning, design, implementation, testing, and documentation.

Task	Start Date	End Date	Duration	W1	W2	W3	W4	W5	W6	W7	W8
Project Planning	February 2, 2026	February 6, 2026	5 days	X							
Database Design (ERD, migrations)	February 9, 2026	February 13, 2026	5 days		X						
Wireframing and UI/UX	February 16, 2026	February 20, 2026	5 days			X					
Backend Development (Models, Controllers, Routes)	February 23, 2026	March 6, 2026	10 days				X	X			
Frontend Development (Blade, CSS, JS)	March 2, 2026	March 13, 2026	10 days					X	X		
Testing and Debugging	March 16, 2026	March 20, 2026	5 days							X	
Documentation and Finalization	March 23, 2026	March 27, 2026	5 days								X

IX. Entity-Relationship Diagram (ERD)

The Farm Supply Inventory System uses a relational structure that links users, product data, inventory, stock movement, orders, sales, reviews, notifications, and governance records. The simplified ERD below summarizes the major entities used by the application.

```
ROLES 1 ---- * USERS
BRANCHES 1 ---- * WAREHOUSES
```

```
CATEGORIES 1 ---- * PRODUCTS
SUPPLIERS 1 ---- * PRODUCTS
SUPPLIERS 1 ---- * INVENTORY
PRODUCTS 1 ---- * INVENTORY
WAREHOUSES 1 ---- * INVENTORY
PRODUCTS 1 ---- * TRANSACTIONS
INVENTORY 1 ---- * TRANSACTIONS
USERS 1 ---- * TRANSACTIONS

USERS 1 ---- * ORDERS
ORDERS 1 ---- * ORDER_ITEMS
PRODUCTS 1 ---- * ORDER_ITEMS
USERS 1 ---- * USER_NOTIFICATIONS
USERS 1 ---- * PRODUCT_REVIEWS
PRODUCTS 1 ---- * PRODUCT_REVIEWS
USERS 1 ---- * CUSTOMER_REQUESTS
PRODUCTS 1 ---- * CUSTOMER_REQUESTS

USERS 1 ---- * SALES
SALES 1 ---- * SALE_ITEMS
PRODUCTS 1 ---- * SALE_ITEMS
```

Entity	Sample Attributes
Roles	id, role_name
Users	id, role_id, name, email, phone, address, status
Categories	id, slug, name, description, is_active
Suppliers	id, name, contact_person, email, phone
Products	id, category_id, supplier_id, sku, name, unit_price, current_stock, reorder_point
Branches	id, code, name, location, is_active
Warehouses	id, branch_id, code, name, location, is_active
Inventory	id, product_id, warehouse_id, supplier_id, batch_number, quantity, unit_cost, selling_price, expiry_date, status
Transactions	id, product_id, inventory_id, warehouse_id, user_id, transaction_type, quantity, total_amount

Entity	Sample Attributes
Orders	id, customer_user_id, processed_by_user_id, order_number, status, payment_method, total
Order Items	id, order_id, product_id, quantity, unit_price, line_total
Sales	id, customer_user_id, cashier_user_id, sale_number, payment_method, discount_type, total
Sale Items	id, sale_id, product_id, quantity, unit_price, line_total
User Notifications	id, user_id, type, title, message, data

Full editable ERD source: **docs/Farm_Supply_Inventory_System_ERD_Labeled.drawio**

Relationship notes used for this chapter: **docs/Farm_Supply_Inventory_System_ERD.md**